

HOSPITAL APPOINTMENT WAITING TIME DATASET

EDA AND REGULARIZATION REPORT

Dataset Shape: (100000, 12)

Target Variable: waiting_time

Numerical Features: 2

Categorical Features: 9

Missing Values: 0

Duplicate Rows: 4

MODEL PERFORMANCE

Linear Regression

MAE: 6.4261
MSE: 64.2987
RMSE: 8.0186
R2 Score: 0.9295

Ridge Regression

MAE: 6.4261
MSE: 64.2984
RMSE: 8.0186
R2 Score: 0.9295

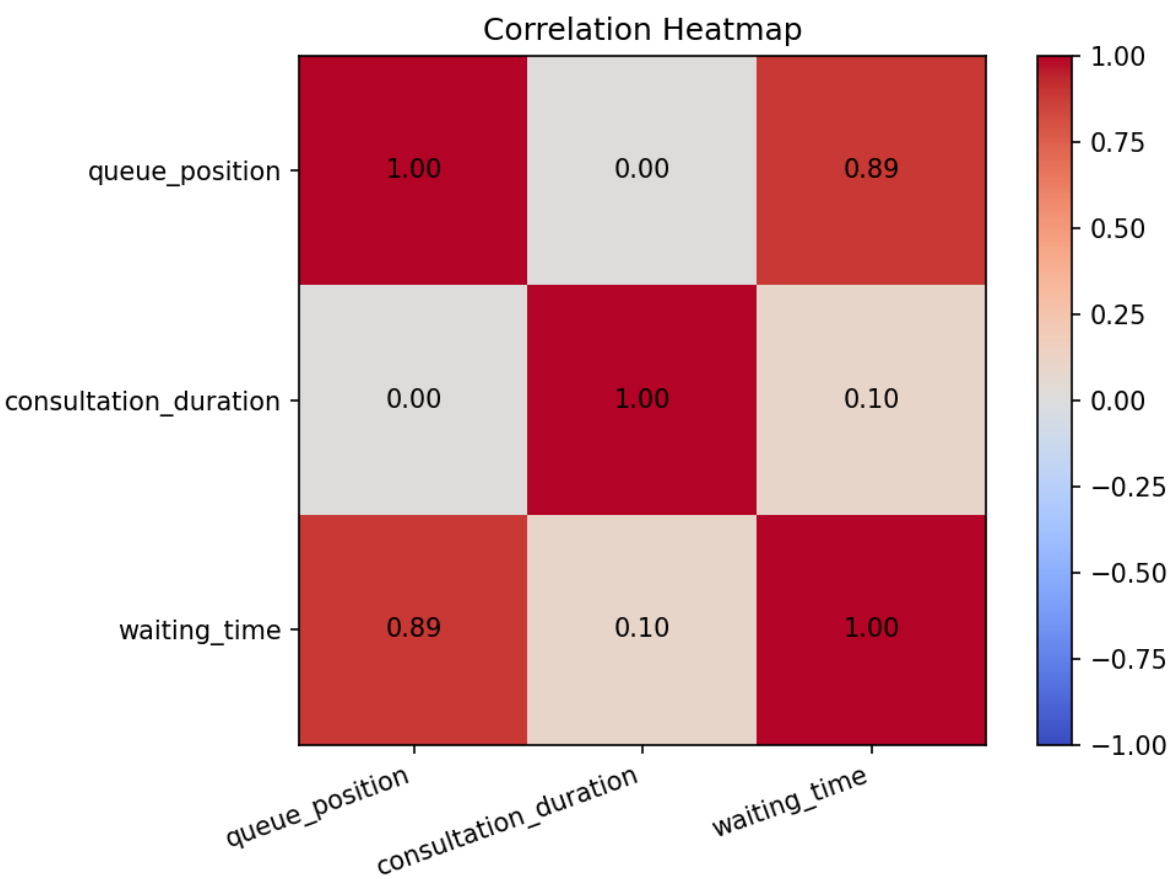
Lasso Regression

MAE: 6.4263
MSE: 64.3038
RMSE: 8.0190
R2 Score: 0.9295

Elastic Net

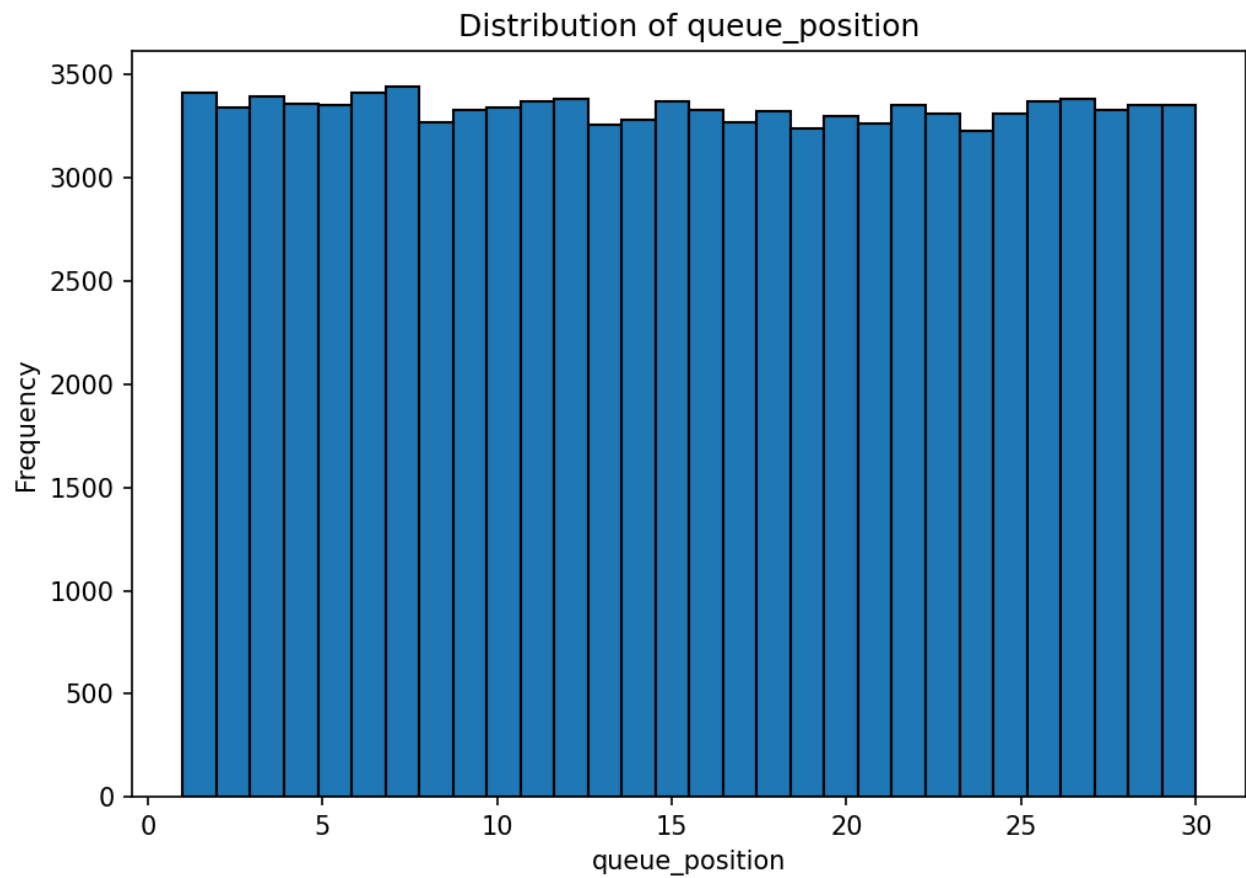
MAE: 6.4359
MSE: 64.4676
RMSE: 8.0292
R2 Score: 0.9293

Correlation Heatmap



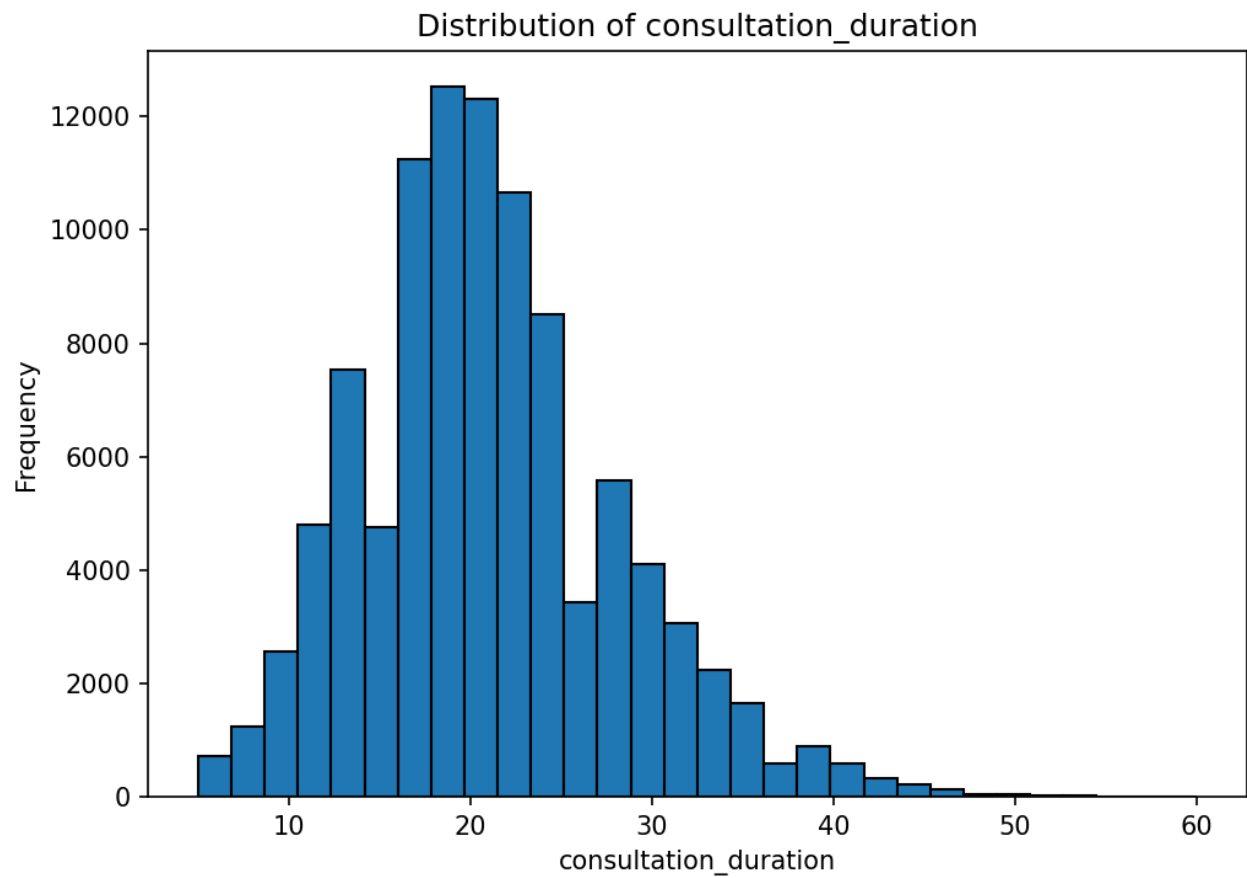
The heatmap shows the relationships among queue_position, consultation_duration and waiting_time.

Distribution of queue_position



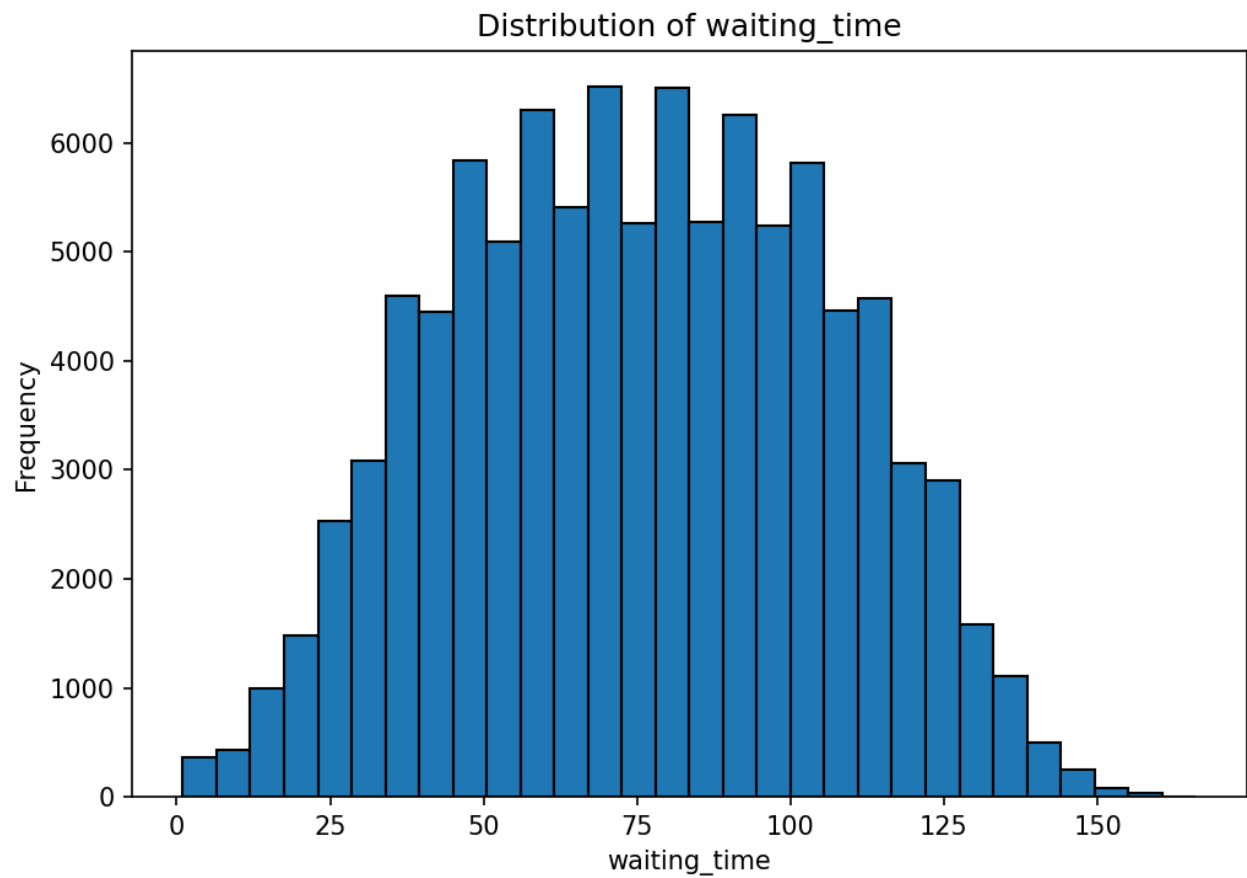
The distribution shows the frequency of patients at different queue positions.

Distribution of consultation_duration



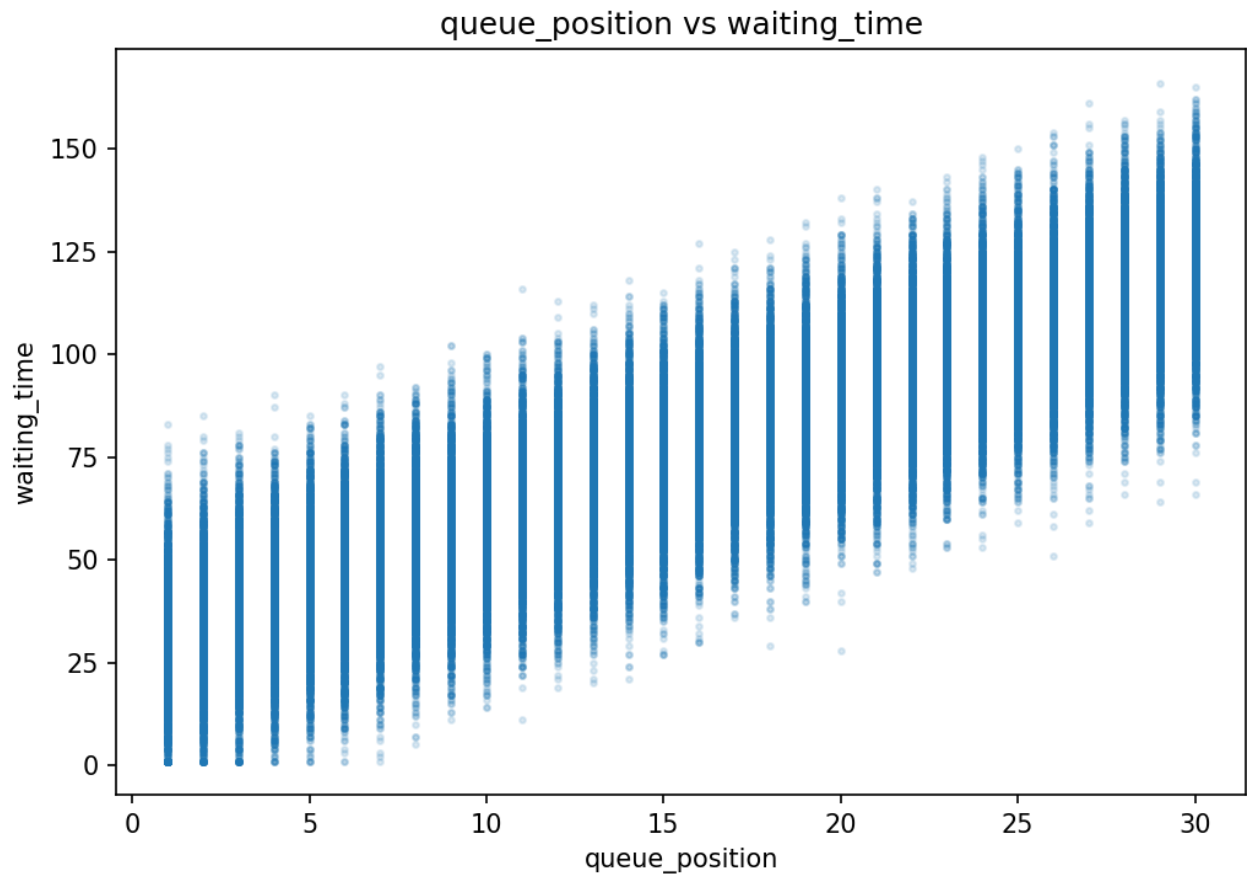
The distribution shows the variation in consultation duration.

Distribution of waiting_time



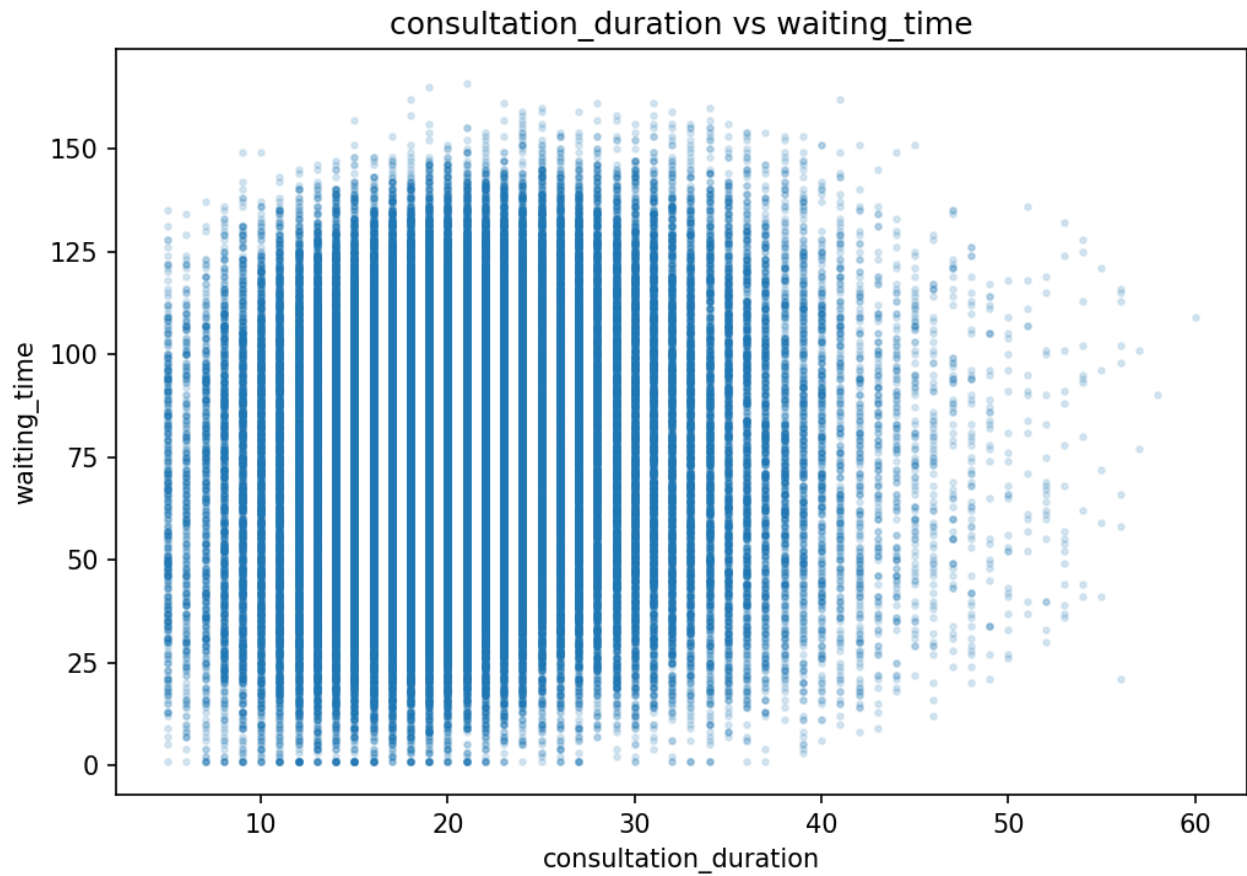
The distribution shows the frequency of hospital appointment waiting times.

queue_position vs waiting_time



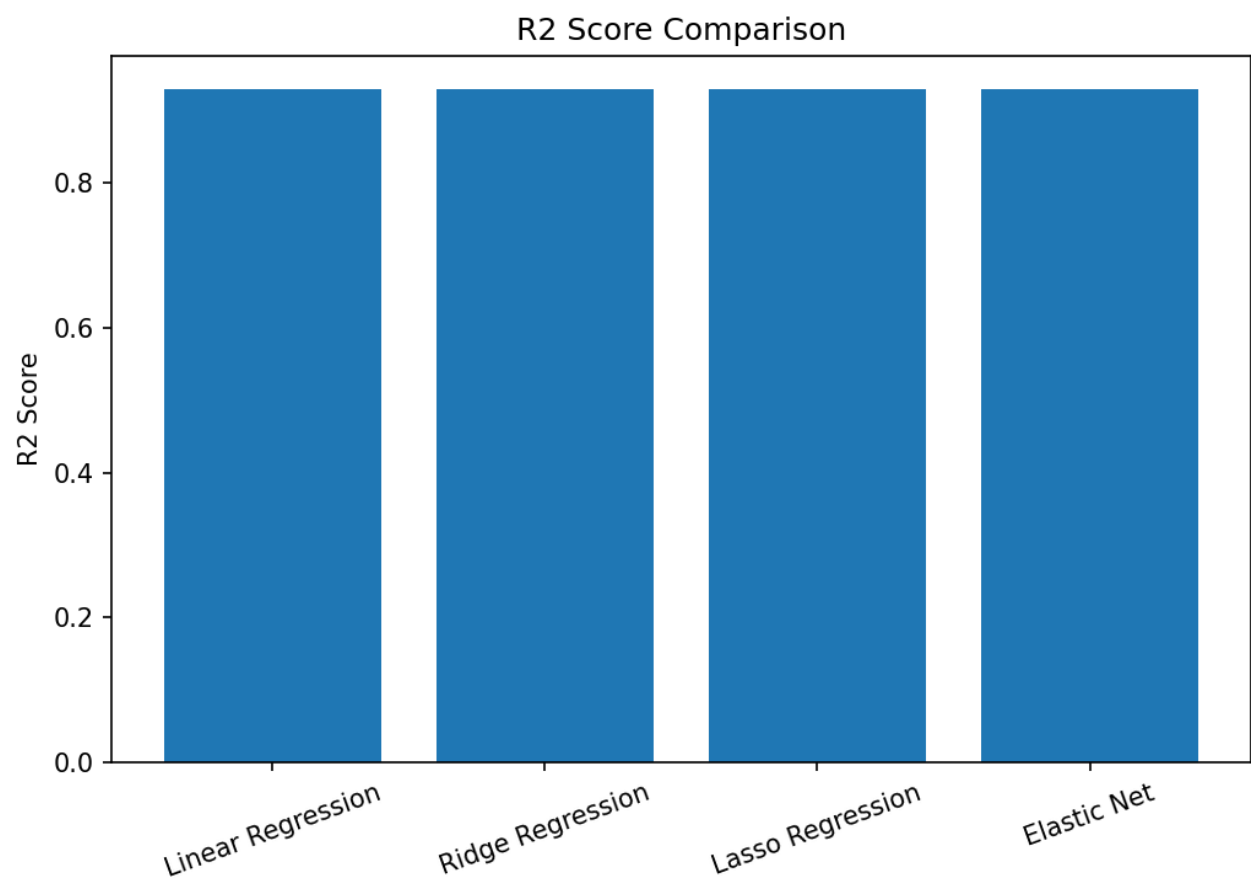
Patients with higher queue positions generally have greater waiting times.

consultation_duration vs waiting_time



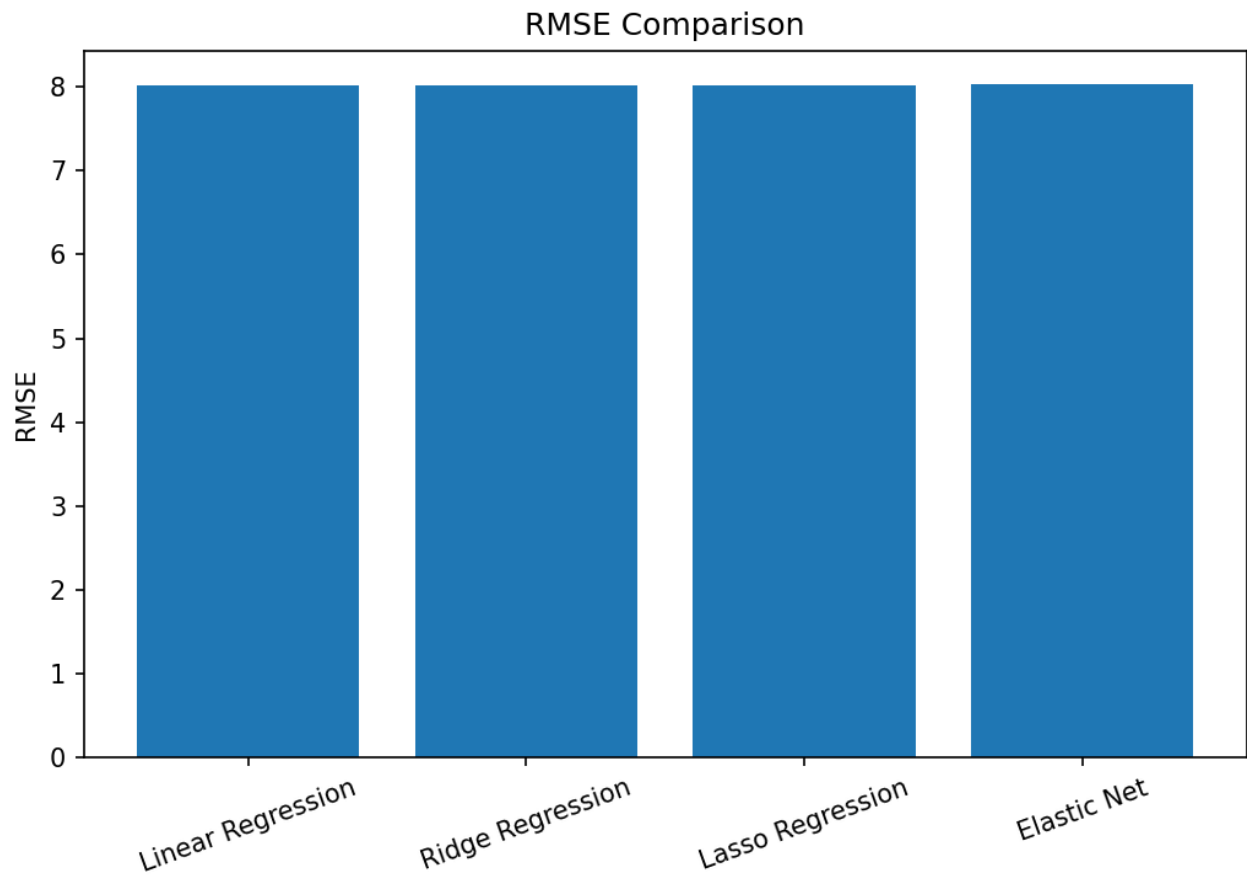
Consultation duration contributes to variation in waiting time.

R2 Score Comparison



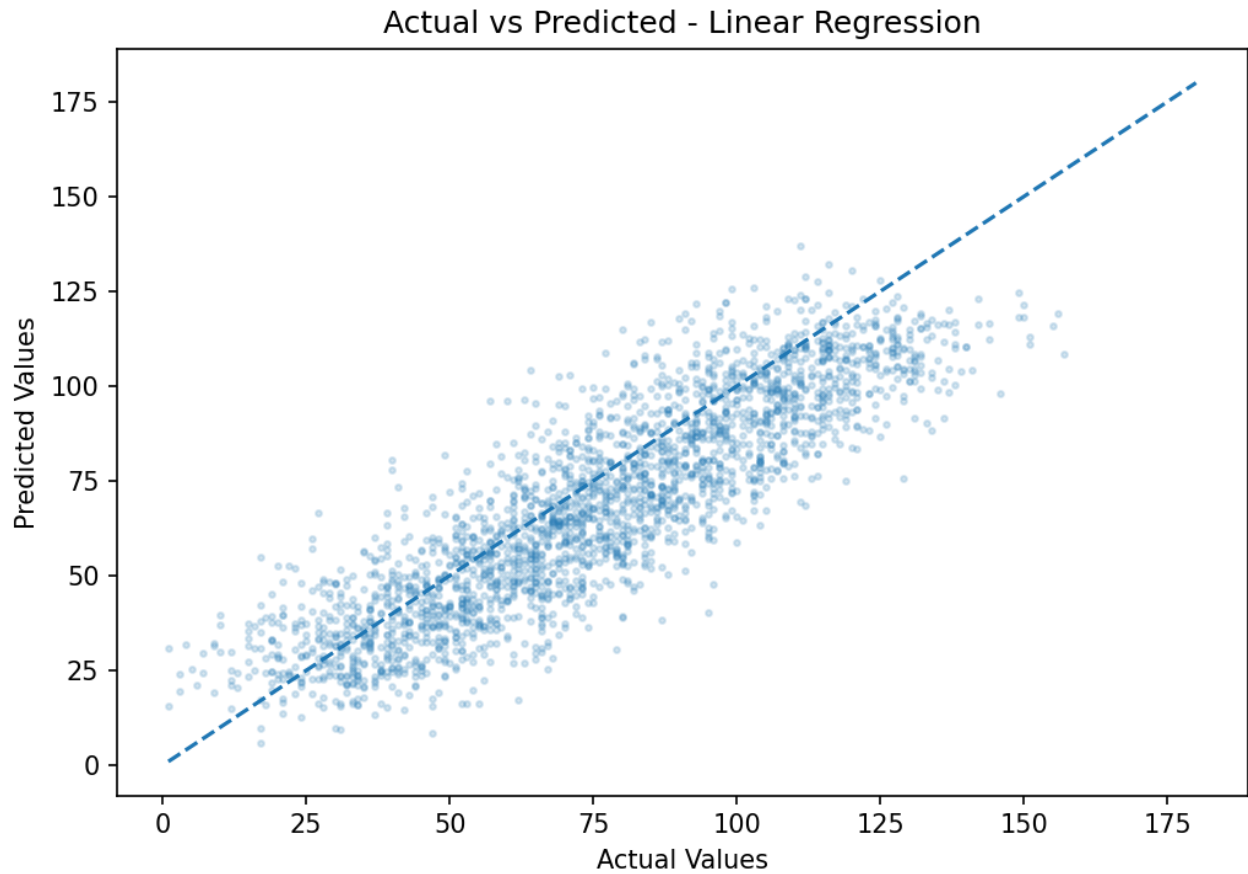
The R2 comparison shows the predictive performance of the four regression models.

RMSE Comparison



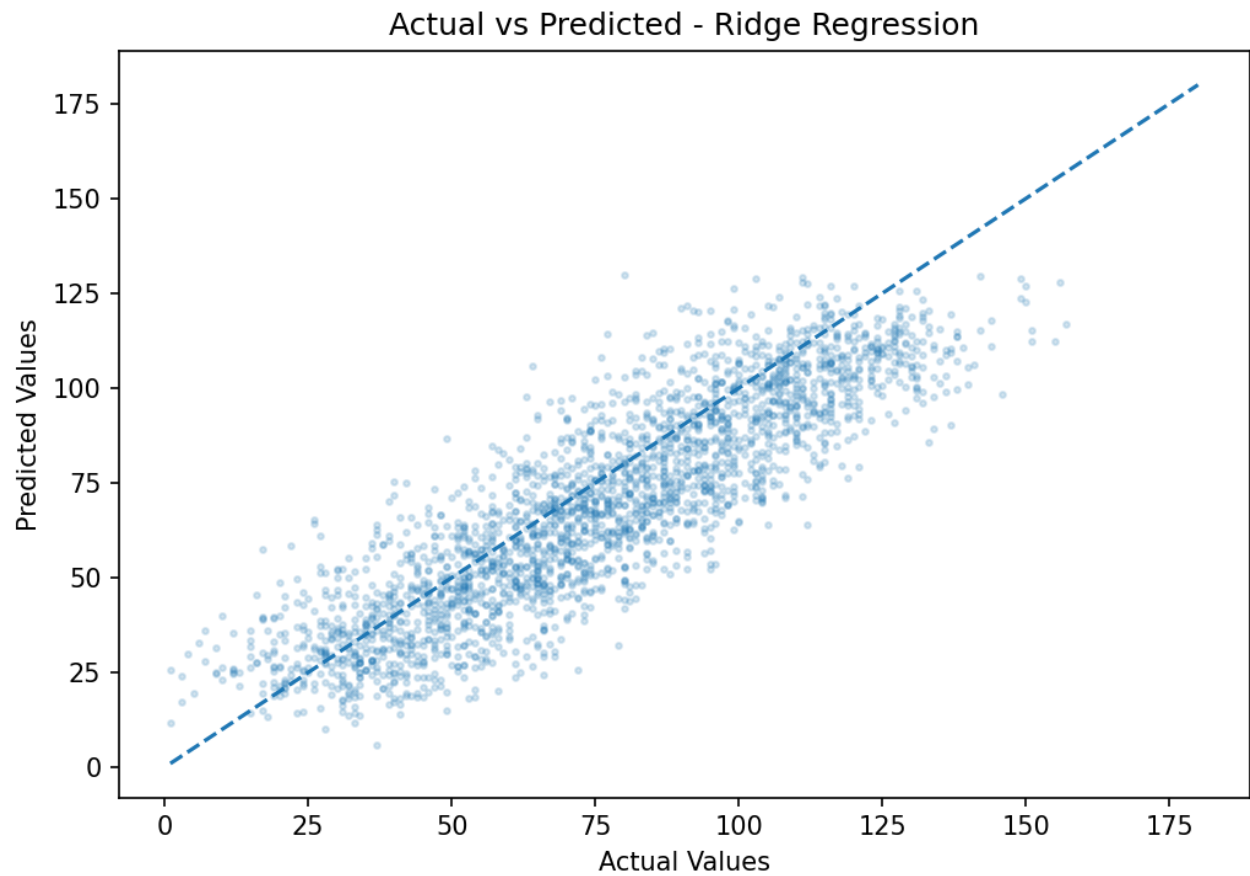
The RMSE comparison shows the prediction error of the four regression models.

Actual vs Predicted - Linear Regression



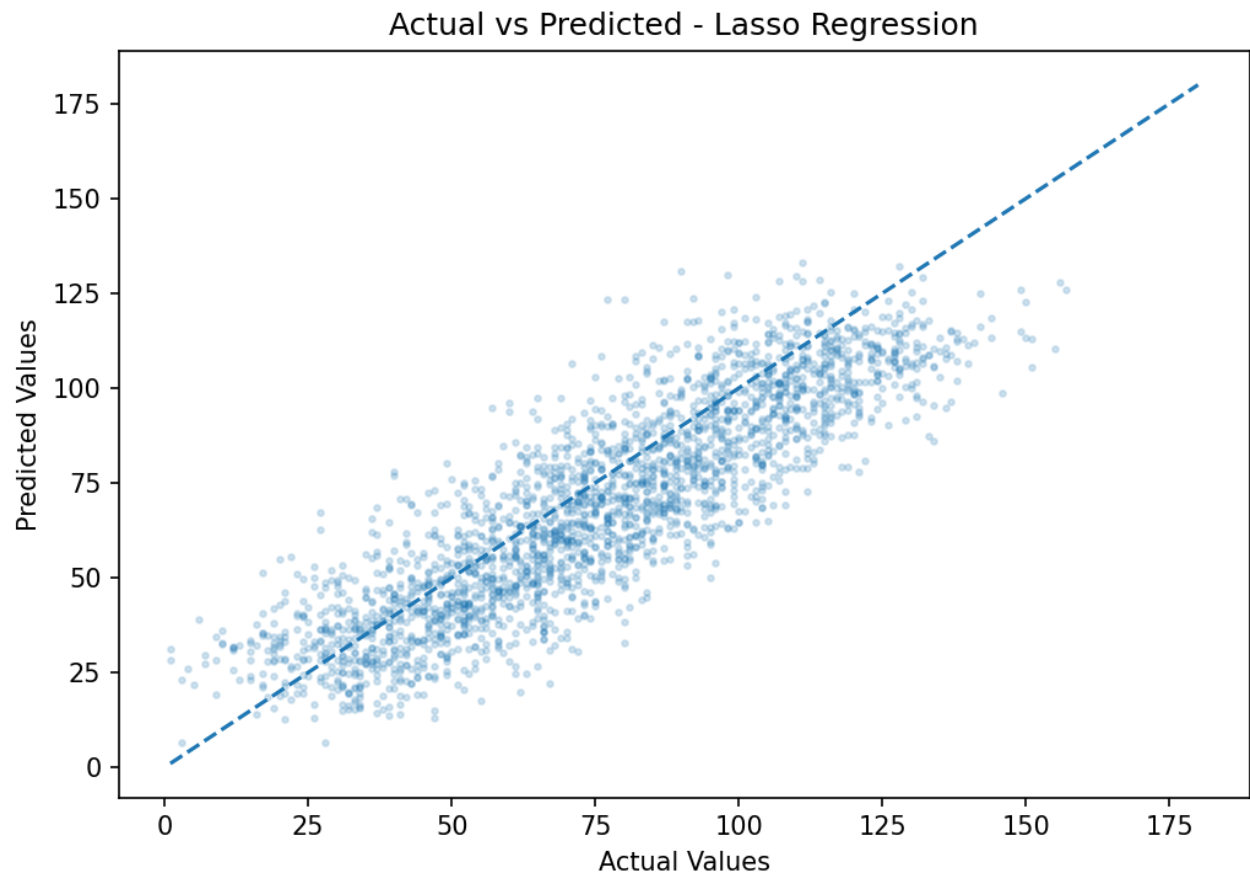
Actual and predicted waiting-time values are compared for Linear Regression.

Actual vs Predicted - Ridge Regression



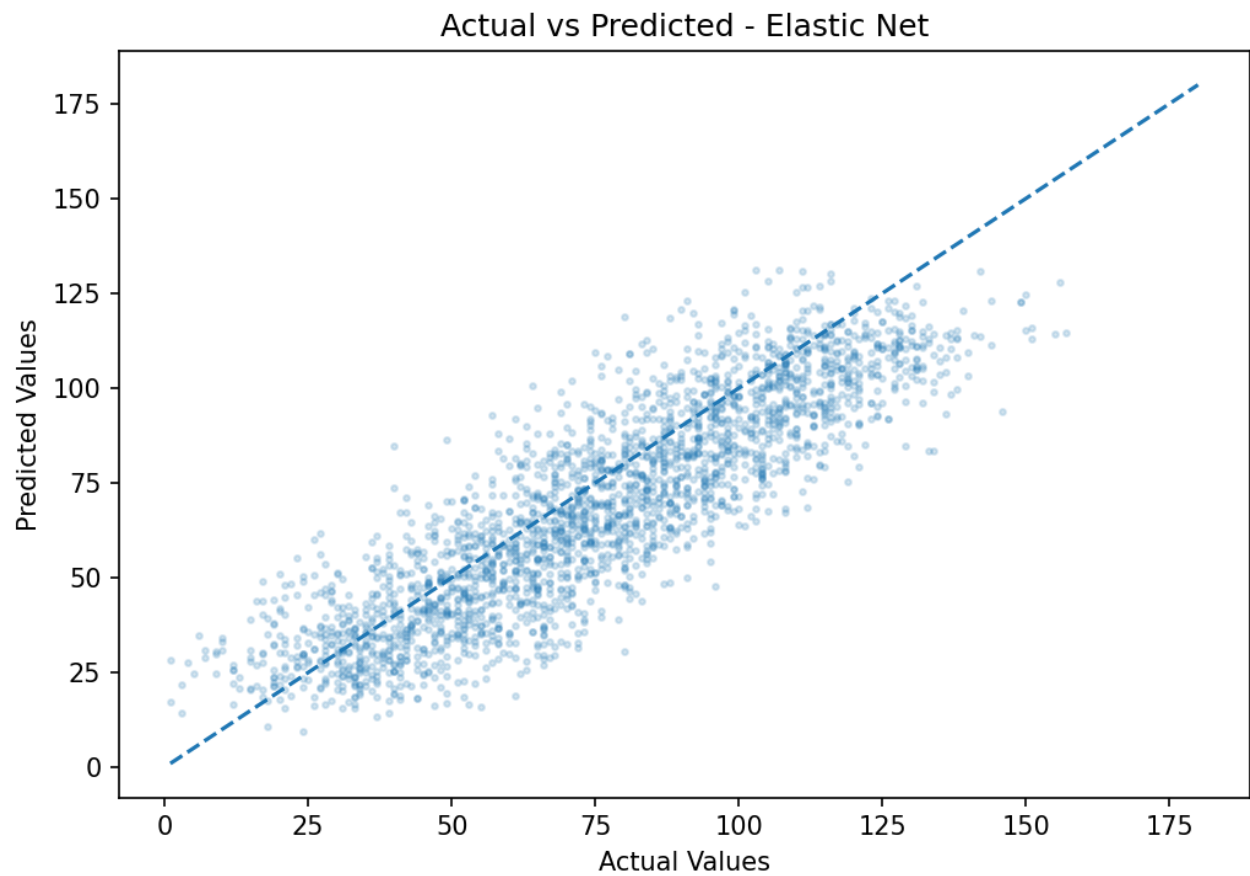
Actual and predicted waiting-time values are compared for Ridge Regression.

Actual vs Predicted - Lasso Regression



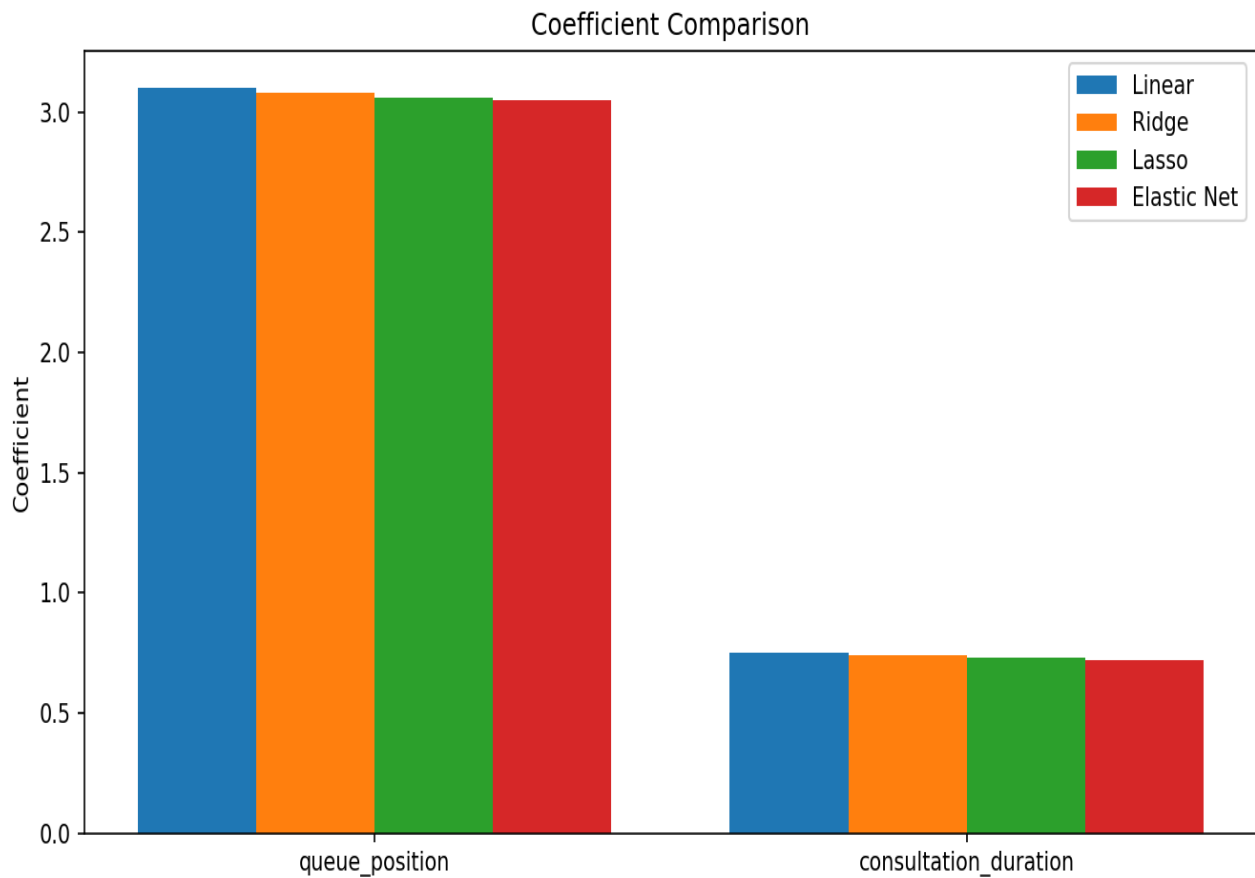
Actual and predicted waiting-time values are compared for Lasso Regression.

Actual vs Predicted - Elastic Net



Actual and predicted waiting-time values are compared for Elastic Net.

Coefficient Comparison



The coefficient comparison shows the effect of regularization on feature coefficients.

FINAL OBSERVATIONS

Best Model: Ridge Regression

Best R2 Score: 0.9295

Best RMSE: 8.0186

Lasso Zero Coefficients: 17

Elastic Net Zero Coefficients: 7

The Hospital Appointment Waiting Time Synthetic Dataset was successfully loaded and explored. The dataset contains 100000 records and 12 attributes, with 2 numerical features and 9 categorical features. No missing values were found and 4 duplicate rows were identified.

The categorical features were encoded using One-Hot Encoding, while numerical features were standardized using StandardScaler. The dataset was divided into training and testing sets using an 80:20 split.

Linear Regression was used as the baseline model. Ridge Regression was implemented using L2 regularization, Lasso Regression using L1 regularization, and Elastic Net using a combination of L1 and L2 regularization.

The models were evaluated using MAE, MSE, RMSE and R2 Score. Ridge Regression achieved the highest R2 Score of 0.9295 and the lowest RMSE of 8.0186, making it the best-performing model for this experiment.

Ridge Regression applies L2 regularization, Lasso Regression applies L1 regularization, and Elastic Net combines L1 and L2 regularization. The effect of regularization on prediction performance and model coefficients was successfully observed.